

EECS 6720 Homework 1

Jihyun Park

Problem 1

Problem 1

Goal is to maximize $\ln p(R, U, V) = \ln \frac{p(R, U, V, \phi)}{p(\phi | R, U, V)}$

since $\ln p(R, U, V)$ does not depend on ϕ ,

$$\int q(\phi) \ln p(R, U, V) d\phi = \int q(\phi) \ln \left(\frac{p(R, U, V, \phi)}{q(\phi)} \frac{q(\phi)}{p(\phi | R, U, V)} \right)$$

$$\therefore \ln p(R, U, V) = \underbrace{\int q(\phi) \ln \frac{p(R, U, V, \phi)}{q(\phi)} d\phi}_{L(U, V)} + \underbrace{\int q(\phi) \ln \frac{q(\phi)}{p(\phi | R, U, V)} d\phi}_{D_{KL}(q \| p)}$$

(1a) Derive $q(\phi)$ (Show $q(\phi) = \prod_{(i,j) \in \mathcal{R}} q(\phi_{ij})$)

To maximize $\ln p(R, U, V)$, we can maximize $L(U, V)$ by making $D_{KL}(q \| p) = 0$ by setting $q(\phi) = p(\phi | R, U, V)$

since $r_{ij} \stackrel{\text{ind}}{\sim} \text{Bernoulli}$ and $u_i \stackrel{\text{iid}}{\sim} N(0, cI)$, $v_j \stackrel{\text{iid}}{\sim} N(0, cI)$,
by Bayes rule, $p(\phi_{ij} | r_{ij}, u_i, v_j) \propto \underbrace{p(r_{ij} | \phi_{ij})}_{\text{likelihood}} \underbrace{p(\phi_{ij} | u_i, v_j)}_{\text{prior}}$
 $r_{ij} = \text{sign}(\phi_{ij})$ $\phi_{ij} \sim \text{Normal}(u_i^T v_j, \sigma^2)$

This is a Gaussian $\times$ step function, resulting in truncated Gaussian:

$$q(\phi_{ij}) = \frac{1}{z_{ij}} N(\phi_{ij} | u_i^T v_j, \sigma^2) \mathbb{1}(r_{ij} \phi_{ij} \geq 0)$$

$$\text{where } z_{ij} = \begin{cases} \Phi\left(\frac{u_i^T v_j}{\sigma}\right) & , r_{ij} = +1 \\ \Phi\left(\frac{-u_i^T v_j}{\sigma}\right) & , r_{ij} = -1 \end{cases}$$

$$\mathbb{1}(r_{ij} \phi_{ij} \geq 0) = \begin{cases} 1 & , r_{ij} \phi_{ij} \geq 0 \\ 0 & , \text{otherwise} \end{cases}$$

$$\therefore q(\phi) = \prod_{(i,j) \in \mathcal{R}} q(\phi_{ij})$$

(1b) Derive $L(U, V)$.

$$L(U, V) = \int q(\phi) \ln \frac{p(R, U, V, \phi)}{q(\phi)} d\phi,$$

$$\begin{aligned} p(R, U, V, \phi) &= p(U, V) p(R, \phi | U, V) \dots \text{by chain rule} \\ &= p(U) p(V) \cdot p(\phi | U, V) p(R | \phi) \\ &= \prod_i p(u_i) \prod_j p(v_j) \prod_{j \in \Omega} p(r_{ij} | \phi_{ij}) \prod_{j \in \Omega} p(\phi_{ij} | u_i, v_j) \end{aligned}$$

$L(U, V)$ can also be expressed as:

$$L(U, V) = \underbrace{\mathbb{E}_q[\ln p(R, U, V, \phi)] - \mathbb{E}_q[\ln q(\phi)]}_{\text{constant}}.$$

$$\begin{aligned} \mathbb{E}_q[\ln p(R, U, V, \phi)] &= \mathbb{E}_q\left[\ln \left\{ \prod_i p(u_i) \prod_j p(v_j) \prod_{j \in \Omega} p(r_{ij} | \phi_{ij}) \prod_{j \in \Omega} p(\phi_{ij} | u_i, v_j) \right\}\right] \\ &= \underbrace{\sum_i \ln p(u_i)}_{-\frac{1}{2c} u_i^T u_i} + \underbrace{\sum_j \ln p(v_j)}_{-\frac{1}{2c} v_j^T v_j} + \sum_{i,j \in \Omega} \underbrace{\mathbb{E}_q[\ln p(r_{ij} | \phi_{ij})]}_{=0} + \underbrace{\mathbb{E}_q[\ln p(\phi_{ij} | u_i, v_j)]}_{-\frac{1}{2\sigma^2} (\phi_{ij} - u_i^T v_j)^2} \\ &= \underbrace{-\frac{|u_i|^2}{2c}}_{-\sum_i \frac{|u_i|^2}{2c}} - \underbrace{\frac{|v_j|^2}{2c}}_{-\sum_j \frac{|v_j|^2}{2c}} + \sum_{i,j \in \Omega} \mathbb{E}_q\left[-\frac{1}{2\sigma^2} (\phi_{ij} - u_i^T v_j)^2\right] \quad \because \phi_{ij} \sim N(u_i^T v_j, \sigma^2) \end{aligned}$$

$$\therefore L(U, V) = -\sum_i \frac{|u_i|^2}{2c} - \sum_j \frac{|v_j|^2}{2c} - \sum_{i,j \in \Omega} \frac{1}{2\sigma^2} \mathbb{E}_q[(\phi_{ij} - u_i^T v_j)^2] + \text{constant}.$$

(1c) U, V that optimize $L(U, V)$ 

$$L(U, V) = -\sum_i \frac{|u_i|^2}{2c} - \sum_j \frac{|v_j|^2}{2c} - \sum_{i,j \in \Omega} \frac{1}{2\sigma^2} \left\{ \mathbb{E}_q[\phi_{ij}^2] - 2\mathbb{E}_q[\phi_{ij}] (u_i^T v_j) + (u_i^T v_j)^2 \right\} + \text{constant}$$

optimizing individual u_i 's

$$\begin{aligned} \frac{\partial L}{\partial u_i} &= \frac{-u_i}{c} - \frac{1}{2\sigma^2} \sum_{j \in \Omega} (-2v_j \mathbb{E}_q[\phi_{ij}] + \underbrace{2(u_i^T v_j) v_j}_{=2v_j(v_j^T u_i)}) = 0 \\ &= 2(v_j v_j^T) u_i \\ &\because u_i^T v_j \text{ is scalar,} \end{aligned}$$

solve for u_i

$$\frac{\partial L}{\partial u_i} = -\frac{u_i}{c} - \frac{1}{2\sigma^2} \sum_{j \in \mathcal{R}} (-2v_j \mathbb{E}_q[\phi_{ij}] + 2(v_j v_j^T) u_i) = 0$$

$$\frac{u_i}{c} + \frac{1}{\sigma^2} \sum_{j \in \mathcal{R}} (v_j v_j^T) u_i = \frac{1}{\sigma^2} \sum_{j \in \mathcal{R}} v_j \mathbb{E}_q[\phi_{ij}]$$

$$= \left(\frac{1}{c} I + \frac{1}{\sigma^2} \sum_{j \in \mathcal{R}} (v_j v_j^T) \right) u_i$$

$$\therefore u_i = \left\{ \frac{1}{c} I + \frac{1}{\sigma^2} \sum_{j \in \mathcal{R}} (v_j v_j^T) \right\}^{-1} \frac{1}{\sigma^2} \sum_{j \in \mathcal{R}} v_j \mathbb{E}_q[\phi_{ij}]$$

$$v_j = \left\{ \frac{1}{c} I + \frac{1}{\sigma^2} \sum_{i \in \mathcal{R}} (u_i u_i^T) \right\}^{-1} \frac{1}{\sigma^2} \sum_{i \in \mathcal{R}} u_i \mathbb{E}_q[\phi_{ij}]$$

($\because v_j$ is identical in form)

(1d) Summary of EM algorithm

① initialize

$$u_i \sim N(0, cI)$$

$$v_j \sim N(0, cI)$$

where c and σ^2 are small valued hyperparameters > 0 .

② E-step

since $q(\phi_{ij})$ is a truncated Gaussian,

$$\text{if } r=1, \quad \phi_{ij} = u_i^T v_j + \sigma \frac{\phi\left(\frac{-u_i^T v_j}{\sigma}\right)}{1 - \Phi\left(\frac{-u_i^T v_j}{\sigma}\right)}$$

$$\text{if } r=-1, \quad \phi_{ij} = u_i^T v_j - \sigma \frac{\phi\left(\frac{-u_i^T v_j}{\sigma}\right)}{\Phi\left(\frac{-u_i^T v_j}{\sigma}\right)}$$

3 M-step

$$\text{update } u_i \leftarrow \left(\frac{\sigma^2}{c} I + \sum_{j \in \mathcal{R}} v_j v_j^T \right)^{-1} \sum_{j \in \mathcal{R}} v_j \phi_{ij}$$

$$v_j \leftarrow \left(\frac{\sigma^2}{c} I + \sum_{i \in \mathcal{R}} u_i u_i^T \right)^{-1} \sum_{i \in \mathcal{R}} u_i \phi_{ij}$$

④ Marginal log-likelihood

$$p(R, U, V) = \int p(R, U, V, \phi) d\phi$$

$$= p(U) p(V) \int p(R | \phi) p(\phi | U, V) d\phi$$

if $r = 1$

$$p(R, U, V) = \int_0^\infty \mathcal{N}(\phi_{ij} | u_i^T v_j, \sigma^2) d\phi = \Phi\left(\frac{u_i^T v_j}{\sigma}\right)$$

if $r = -1$

$$p(R, U, V) = \int_{-\infty}^0 \mathcal{N}(\phi_{ij} | u_i^T v_j, \sigma^2) d\phi = \Phi\left(\frac{u_i^T v_j}{\sigma}\right)$$

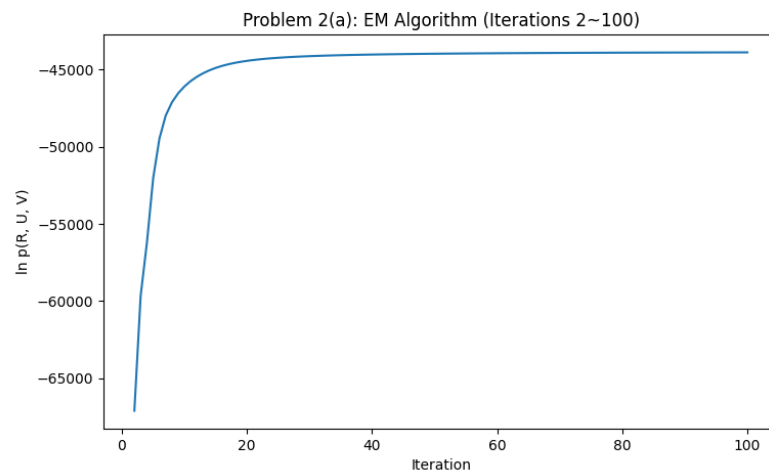
take $\ln$:

$$\ln p(R, U, V) = \sum_{ij \in \mathcal{R}} \ln \Phi\left(\frac{u_i^T v_j r_{ij}}{\sigma}\right) - \sum_i \frac{\|u_i\|^2}{2c} - \sum_j \frac{\|v_j\|^2}{2c} + \text{constant}.$$

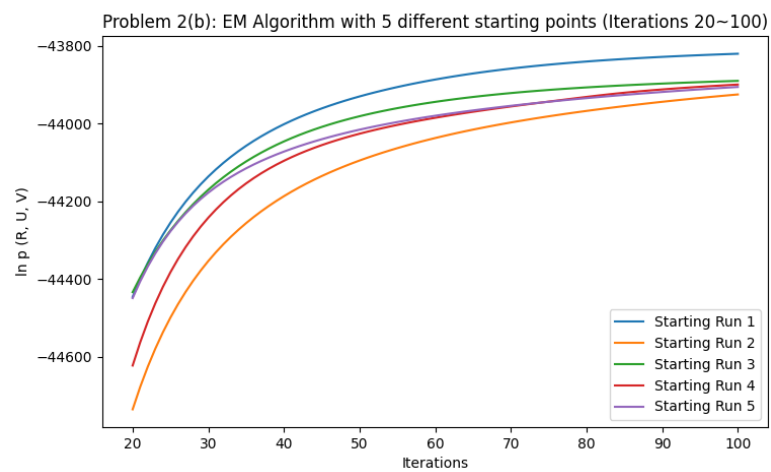
⑤ repeat till log-likelihood converges.

Problem 2

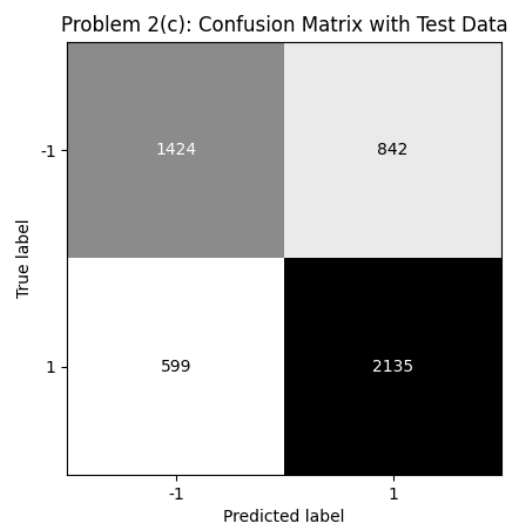
2(a)



2(b)



2(c)



Problem 3

(3a) Log joint likelihood for the model (MCMC sampling approach)

likelihood: $r_{ij} | U, V \stackrel{\text{iid}}{\sim} N(u_i^T v_j, \sigma^2)$, $(i, j) \in \Omega$
conditional density
for a certain observation: $p(r_{ij} | u_i, v_j, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(r_{ij} - u_i^T v_j)^2}{2\sigma^2}\right)$

since u_i and v_j both $\stackrel{\text{iid}}{\sim} N(0, cI)$, for a d -dimensional Gaussian,

$$\begin{aligned} \textcircled{1} \quad p(u_i) &= (2\pi)^{-\frac{d}{2}} |cI_d|^{-\frac{1}{2}} \exp\left(-\frac{1}{2} u_i^T (cI_d)^{-1} u_i\right) \\ &\quad \uparrow \quad \quad \quad \uparrow \\ &\quad (cI_d)^{-1} = c^{-1} I_d \quad (cI_d)^{-1} = \frac{1}{c} I_d \\ &= (2\pi)^{-\frac{d}{2}} c^{-\frac{d}{2}} \exp\left(-\frac{\|u_i\|^2}{2c}\right) \end{aligned}$$

and similarly, $p(v_j) = (2\pi)^{-\frac{d}{2}} c^{-\frac{d}{2}} \exp\left(-\frac{\|v_j\|^2}{2c}\right)$

② full joint density $p(R, U, V)$

$$p(R, U, V) = \underbrace{\left[\prod_{(i,j) \in \Omega} p(r_{ij} | u_i, v_j, \sigma^2) \right]}_{\text{likelihood}} \underbrace{\left[\prod_{i=1}^N p(u_i) \right] \left[\prod_{j=1}^M p(v_j) \right]}_{\text{priors}}$$

now plug-in the densities from ①

$$\begin{aligned} &= \prod_{(i,j) \in \Omega} \left[(2\pi\sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{(r_{ij} - u_i^T v_j)^2}{2\sigma^2}\right) \right] \\ &\times \prod_{i=1}^N \left[(2\pi)^{-\frac{d}{2}} c^{-\frac{d}{2}} \exp\left(-\frac{\|u_i\|^2}{2c}\right) \right] \times \prod_{j=1}^M \left[(2\pi)^{-\frac{d}{2}} c^{-\frac{d}{2}} \exp\left(-\frac{\|v_j\|^2}{2c}\right) \right] \end{aligned}$$

③ Take logs

the likelihood part: $\sum_{(i,j) \in \Omega} \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(r_{ij} - u_i^T v_j)^2}{2\sigma^2} \right]$

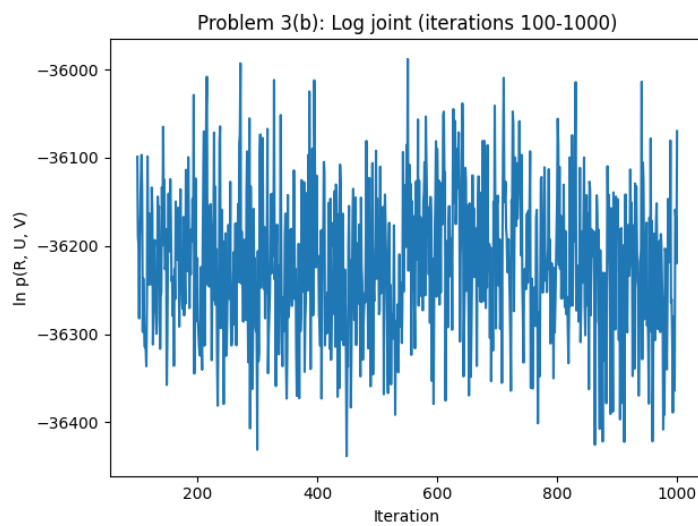
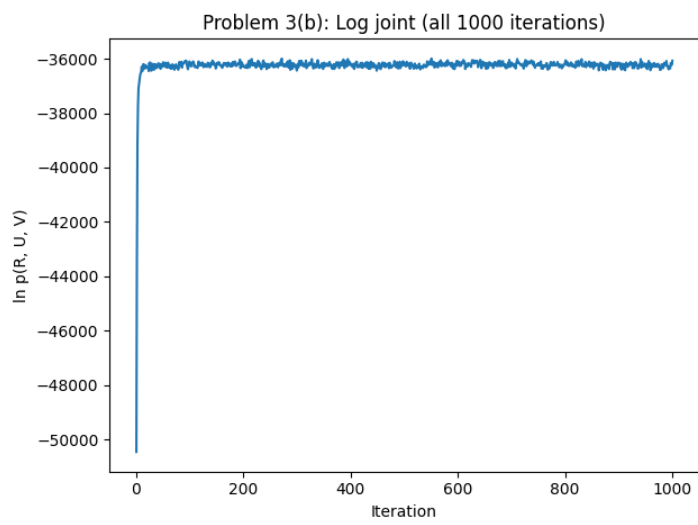
the priors parts: $\sum_{i=1}^N \left[-\frac{d}{2} \log(2\pi) - \frac{d}{2} \log c - \frac{\|u_i\|^2}{2c} \right]$

similarly for v : $\sum_{j=1}^M \left[-\frac{d}{2} \log(2\pi) - \frac{d}{2} \log c - \frac{\|v_j\|^2}{2c} \right]$

④ Combine to get the final log joint likelihood

$$\log p(R, U, V) = -\frac{1}{2\sigma^2} \sum_{(i,j) \in \Omega} (r_{ij} - u_i^T v_j)^2 - \frac{1}{2c} \sum_{i=1}^N \|u_i\|^2 - \frac{1}{2c} \sum_{j=1}^M \|v_j\|^2 + \text{constant}$$

3(b)



3(c)

